

 Settings

Language

English



Theme Color

blue

 Stats

Messages

0

Sessions

1

 Clear All Data

Streamlit Tutorial Demo

Learn Core Concepts with Interactive Examples

>  Current Configuration

Session State: Conversation History

Messages persist across reruns (script restarts). Try clicking buttons multiple times!

Your message:

Hello!

Add Message

No messages yet. Add one!

Clear History



Caching: Load Models & Data

Click buttons below. First click is slow (loading), subsequent clicks are fast (cached).

Load Model (cache_resource)

Fetch Data (cache_data)

▼  View Cache Metrics

@st.cache_resource: One instance per process, shared across reruns

@st.cache_data: One cache per user session

```
@st.cache_resource
def load_expensive_model():
    return SentenceTransformer("all-MiniLM-L6-v2")

@st.cache_data
```

```
def fetch_database():  
    return pandas.read_csv("data.csv")
```



Forms: Multi-Step Input

Forms prevent duplicate submissions and validate input.

Clinical Question Submission

Clinical Question:

Which medicine is typical for high blood pressure?

☒ Include Evidence Snippets

Response Language:

English

Submit Question

Question received and processed!

Question: Which medicine is typical for high blood pressure?

Include Evidence: Yes

Response Language: English



Tabs: Organize Content

Answer **Evidence** Debug Logs

Retrieved Evidence

Source: Clinical Guidelines 2024

- β -blockers can precipitate asthma attacks
- Selective β_1 -blockers may be safer but still require caution
- Alternative: Use calcium channel blockers (e.g., nifedipine)



Advanced: Control Flow

This button demonstrates `st.stop()`

 Rerun Script

Click to Stop Execution



Key Takeaways

1. **Script Reruns:** Streamlit reruns entire script on every user interaction
2. **Session State:** Use `st.session_state` to persist data across reruns
3. **Caching:** Use `@cache_resource` for singletons, `@cache_data` for per-user data
4. **Forms:** Use `st.form()` to group inputs and prevent duplicate submissions
5. **Organization:** Use tabs, expanders, and columns for clean UI
6. **Bilingual:** Dictionary-based translations for multi-language apps
7. **Control Flow:** Use `st.stop()` and `st.rerun()` to control execution



Useful Links

- [Streamlit Docs](#)
- [Session State API](#)
- [Caching](#)
- [CareMind Docs](#)

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